Confusion Matrix

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One labelled confusion-count evidence record.

Trace error types, verify totals and explain the limits of the matrix.

Keep one habit: W1–W3/W5 → selected new-count task → project matrix → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

Scores and Calibration Intuition

My remembered idea, example and correction

Nearest Neighbors

My remembered idea, example and correction

Dataset Cards

My remembered idea, example and correction

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main corridor source V1,V2clear/clear;V3clear/blocked;V4,V5blocked/clear;V6,V7,V8blocked/blocked. Clear/blocked axes: [[2,1],[2,3]],TN2,FP1,FN2,TP3,total8,correct5,62.5%. Separate pictured/lab twenty-leaf source has spotted/healthy axes [[4,1],[3,12]],TP4,FN1,FP3,TN12,total20,correct16,80%. Reviewed correction FN→TP gives [[5,0],[3,12]],17/20reviewed outcomes while original model-only evidence remains unchanged.

  Class order and class definition must stay explicit.

- Reference for this exercise: open this week's student lesson, "Confusion Matrix", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Use all eight worked IDs. Draw a matrix with reference rows and **prediction** columns, both ordered clear then blocked. Place the IDs in cells before writing the counts.

> **My reasoning and evidence**

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main corridor source V1,V2clear/clear;V3clear/blocked;V4,V5blocked/clear;V6,V7,V8blocked/blocked. Clear/blocked axes: [[2,1],[2,3]],TN2,FP1,FN2,TP3,total8,correct5,62.5%. Separate pictured/lab twenty-leaf source has spotted/healthy axes [[4,1],[3,12]],TP4,FN1,FP3,TN12,total20,correct16,80%. Reviewed correction FN→TP gives [[5,0],[3,12]],17/20reviewed outcomes while original model-only evidence remains unchanged.

  Class order and class definition must stay explicit.

- Declare blocked positive. Give TP,TN,FP,FN counts and explain V3 and V4 in plain language. Does positive mean a desirable outcome?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main corridor source V1,V2clear/clear;V3clear/blocked;V4,V5blocked/clear;V6,V7,V8blocked/blocked. Clear/blocked axes: [[2,1],[2,3]],TN2,FP1,FN2,TP3,total8,correct5,62.5%. Separate pictured/lab twenty-leaf source has spotted/healthy axes [[4,1],[3,12]],TP4,FN1,FP3,TN12,total20,correct16,80%. Reviewed correction FN→TP gives [[5,0],[3,12]],17/20reviewed outcomes while original model-only evidence remains unchanged.

  Class order and class definition must stay explicit.

- Reference for this exercise: open this week's student lesson, "Confusion Matrix", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Give the worked row totals, column totals, total count and accuracy. Compare its errors with [[3,0],[3,2]] using the same axes and eight reference records.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main corridor source V1,V2clear/clear;V3clear/blocked;V4,V5blocked/clear;V6,V7,V8blocked/blocked. Clear/blocked axes: [[2,1],[2,3]],TN2,FP1,FN2,TP3,total8,correct5,62.5%. Separate pictured/lab twenty-leaf source has spotted/healthy axes [[4,1],[3,12]],TP4,FN1,FP3,TN12,total20,correct16,80%. Reviewed correction FN→TP gives [[5,0],[3,12]],17/20reviewed outcomes while original model-only evidence remains unchanged.

  Class order and class definition must stay explicit.

- A report gives zero **false positive**s but omits the evaluated sample size, axes and class definitions. What information is missing, and why does zero not prove that future false alarms are impossible?

> **My reasoning and evidence**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Declare actual positive/negative as row order and predicted positive/negative as column order before placing the supplied counts. If you illustrate positive as spotted, label that interpretation explicitly; the generic count task itself does not supply a diagnosis. Use the new20-case totals, not the pictured counts. No row-level IDs or subgroup evidence are provided, so do not invent them.

- On20 cases: **true positive**s5, **false negative**s4, **false positive**s2, **true negative**s9. Build the **confusion matrix** with actual class as rows and predicted class as columns. Compute accuracy.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Publish the Plant Classifier **confusion matrix** with positive-class definition, counts, denominator, and test-set version.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Explain the two error types for Confusion matrices. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Positive=spotted. TP=6,FN=2,FP=3,TN=9. Draw a matrix with actual labels in rows and **prediction**s in columns. Compute total and accuracy.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Positive is spotted. TP=8, FP=2, FN=4, TN=6.
- Build actual-row/predicted-column matrix and total.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Treat fern as positive. Counts are TP=6, FP=2, FN=3 and TN=9. How many examples and how many correct **prediction**s are there?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 15 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Positive=spotted. TP=6,FN=2,FP=3,TN=9. Draw a matrix with actual labels in rows and **prediction**s in columns. Compute total and accuracy.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Each pair below is reference then **prediction**: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows.

  Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Read the **confusion matrix** as four distinct stories about a plant decision.
- Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/15

Positive means spotted

	predicted spotted	predicted healthy
actual spotted	4	1
actual healthy	3	12

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/15

Confusion Matrix

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fill actual spotted row

Actual spotted: [4, 1]. Four caught; one missed.

2. Fill actual healthy row

Actual healthy: [3, 12]. Three false alarms; twelve correctly clear.

3. Compute accuracy

Correct=4+12=16 of total 20, giving80%.

CHANGE ONE THING

A review corrects one missed spotted leaf, moving it from FN to TP.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Read the **confusion matrix** as four distinct stories about a plant decision.
- Which labelled step matches this detail? Correct=4+12=16 of total 20, giving**80%**.
- 1. Compute accuracy
- 2. Fill actual spotted row
- 3. Fill actual healthy row
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compute accuracy Correct=4+12=16 of total 20, giving**80%**. Read the picture in order. Correct=4+12=16 of total 20, giving**80%**.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A review corrects one missed spotted leaf, moving it from FN to TP. Which conclusion is supported by the supplied facts?
- 1. Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%. One decision becomes correct while the total number of evaluated leaves remains20.

Repair and keep the record

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

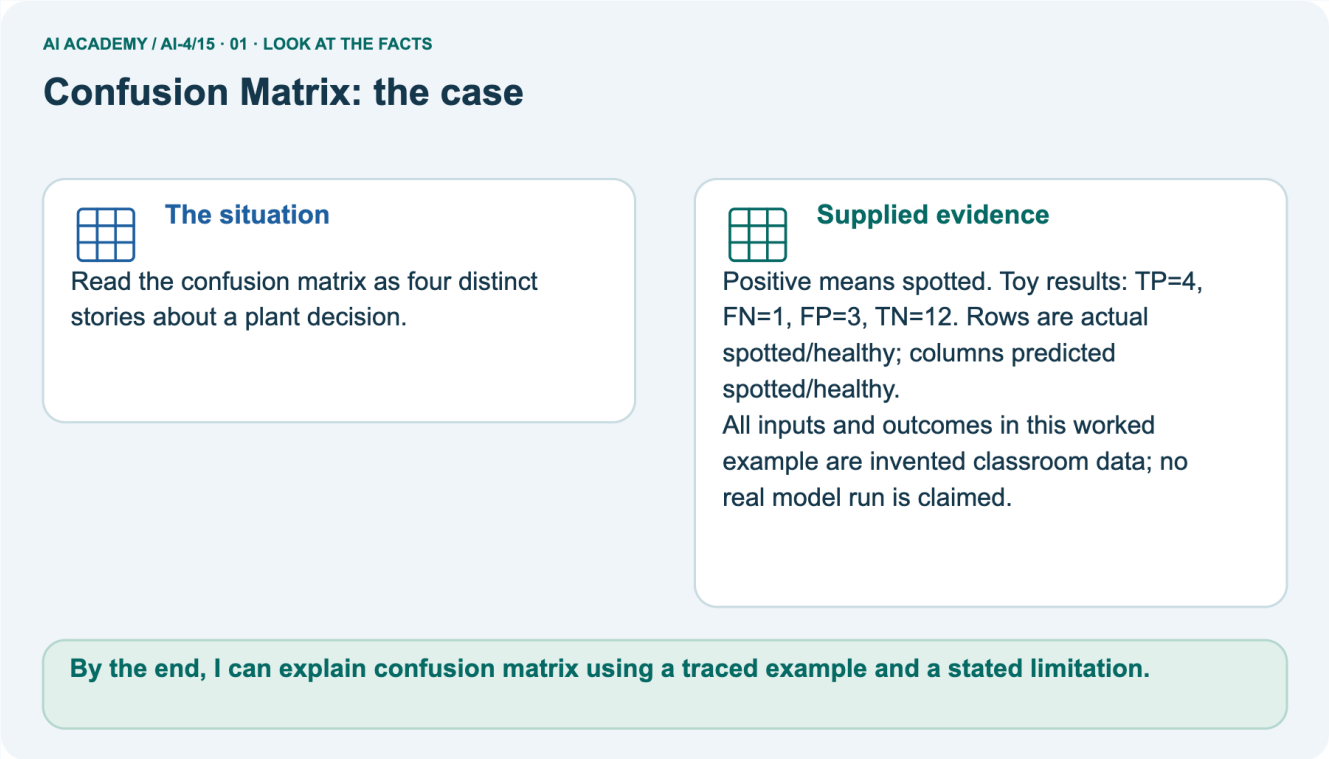
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Confusion Matrix: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

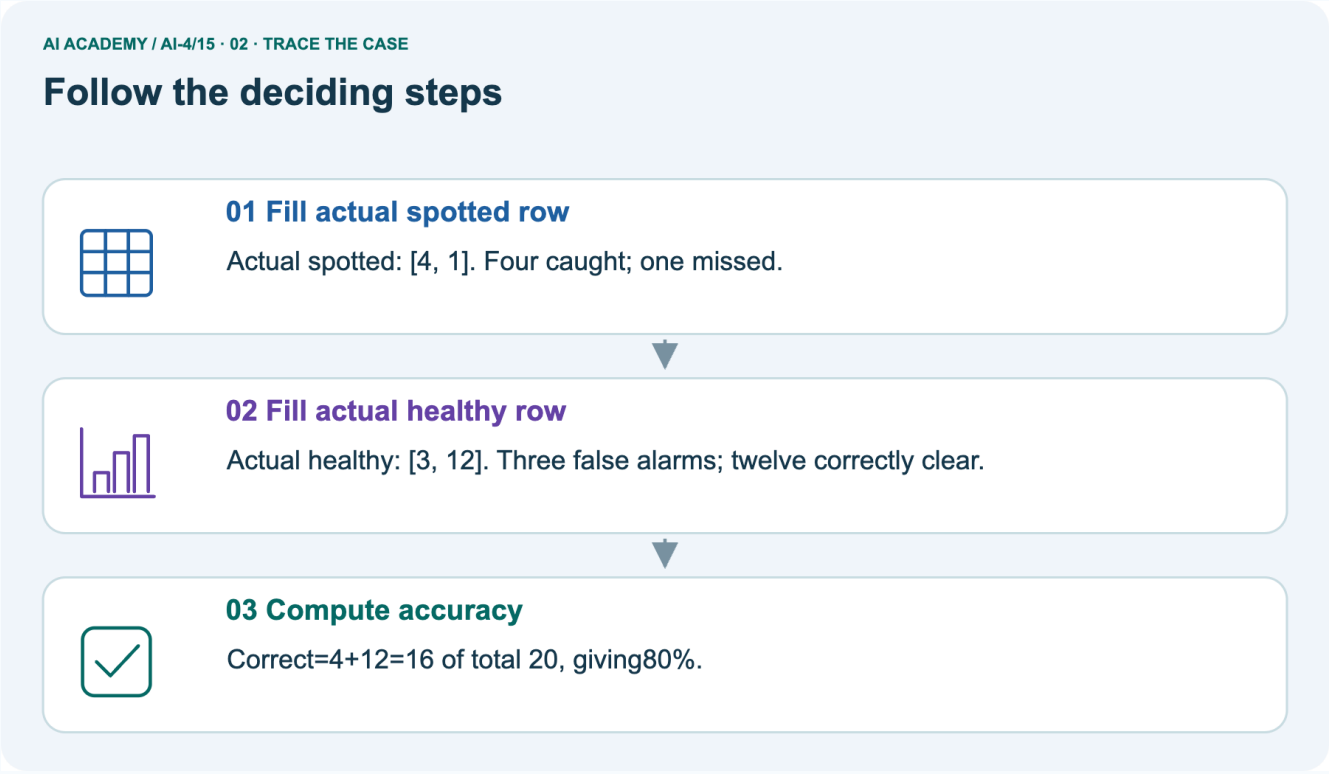
Read the **confusion matrix** as four distinct stories about a plant decision. Supplied facts: Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

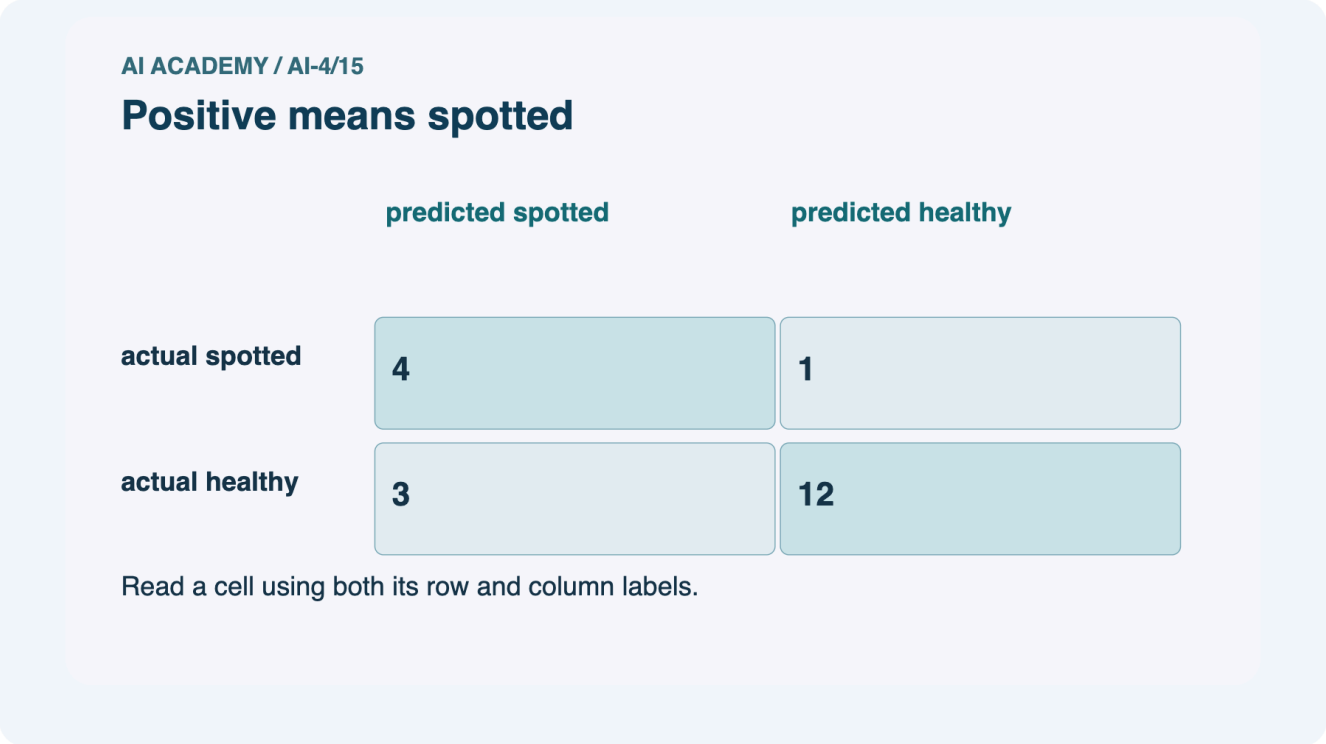
Fill actual spotted row: Actual spotted: [4, 1]. Four caught; one missed. → Fill actual healthy row: Actual healthy: [3, 12].

Three false alarms; twelve correctly clear. → Compute accuracy: Correct=4+12=16 of total 20, giving**80%**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Positive means spotted



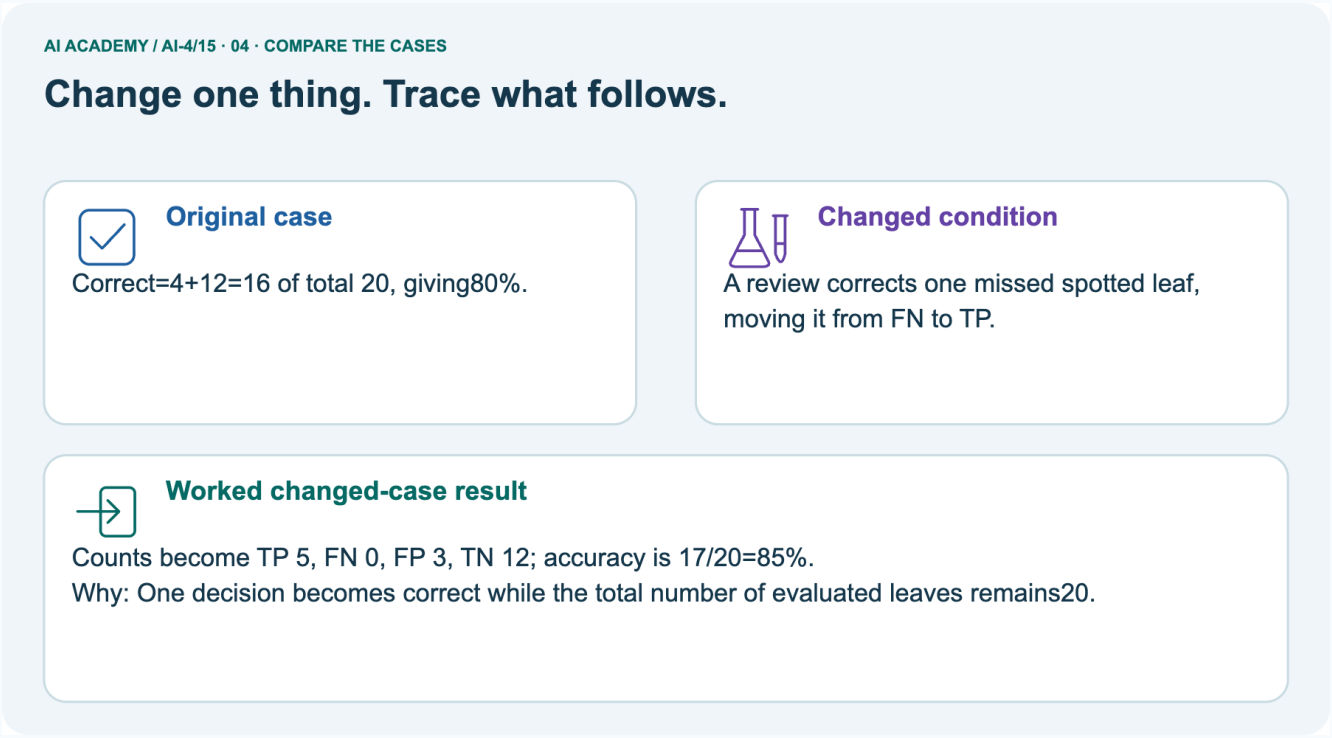
Follow the labelled picture with the matching explanation. Open the vector diagram

Positive means spotted. actual spotted: predicted spotted = 4, predicted healthy = 1. actual healthy: predicted spotted = 3, predicted healthy = 12.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

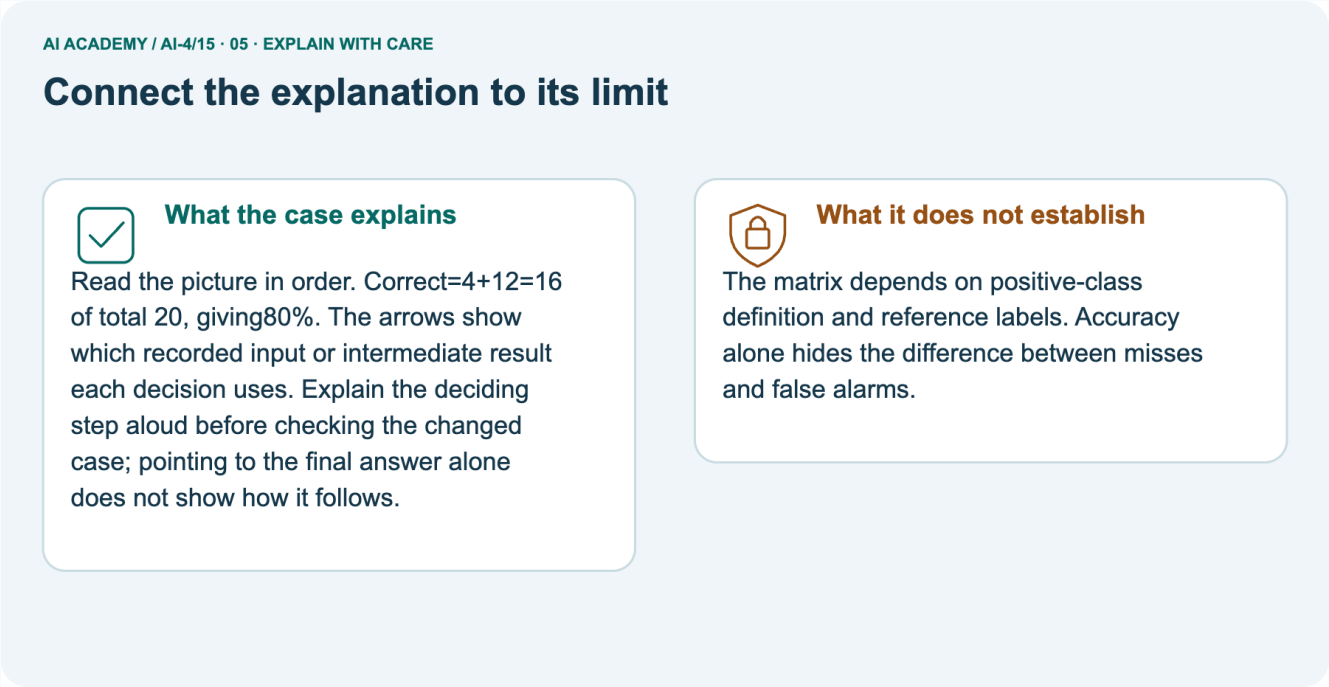
Original case: Correct=4+12=16 of total 20, giving**80%**. Changed condition: A review corrects one missed spotted leaf, moving it from FN to TP. Expected result: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

Reason: One decision becomes correct while the total number of evaluated leaves remains20.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Correct=4+12=16 of total 20, giving**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

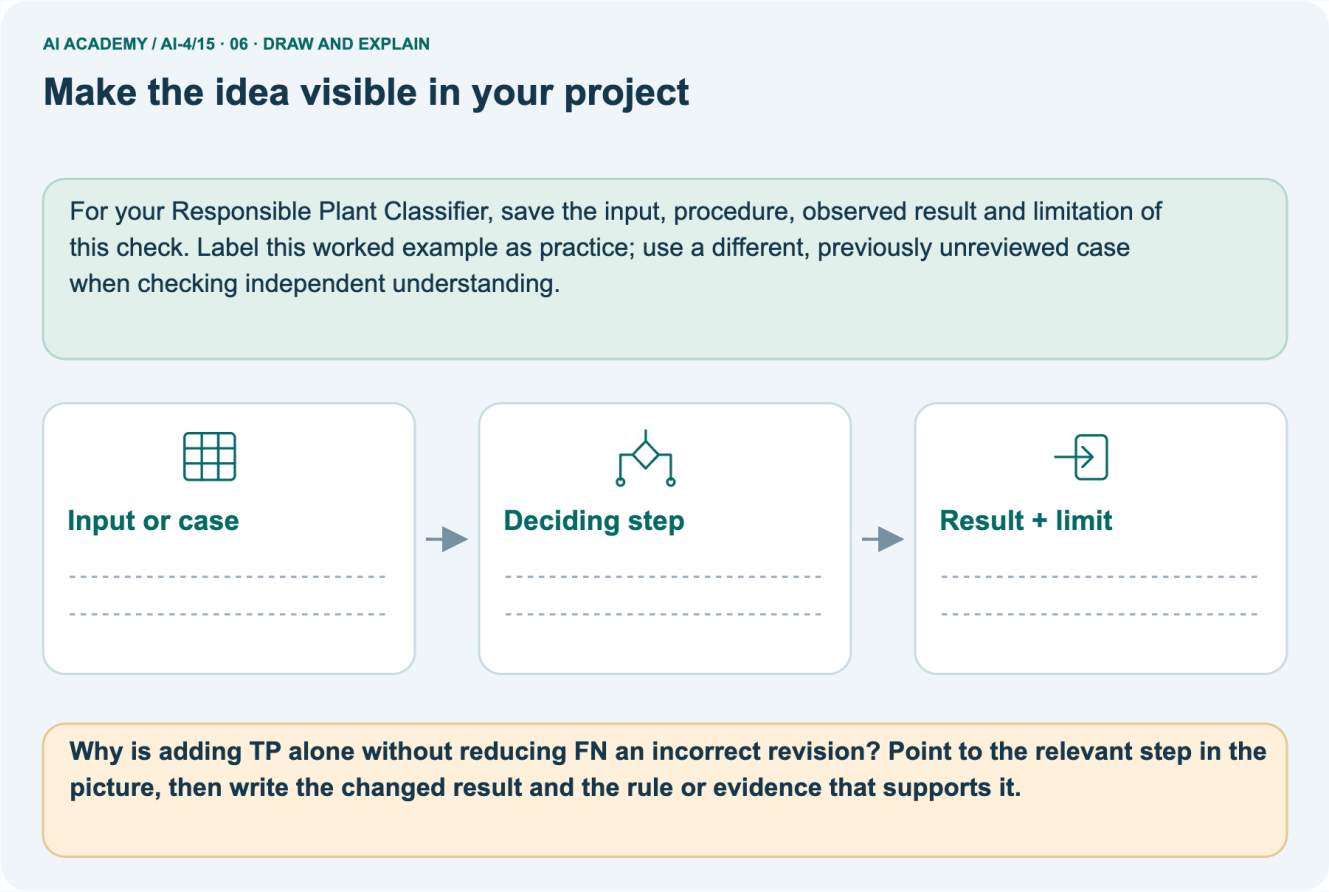
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: The matrix depends on positive-class definition and reference labels.

Accuracy alone hides the difference between misses and false alarms.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Confusion Matrix: the case

02. Follow the deciding steps

03. Positive means spotted

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.